

Chapter 7

Obidi Calculus

Addition, Multiplication, Contraction, and Sectoral Algebra

7.1 Introduction

The Hybrid Manifold Architecture System and the Obidi Convention establish the geometric and notational foundations of the Theory of Entropicity. However, a geometric framework alone is insufficient for constructing a physical theory.

A consistent algebra is required.

General Relativity possesses tensor calculus.

Quantum mechanics possesses operator algebra.

Gauge theories possess Lie algebra structures.

Likewise, the Theory of Entropicity requires a mathematical framework governing interactions among multiple geometric sectors.

This framework is called **Obidi Calculus**.

Obidi Calculus extends conventional tensor algebra to accommodate the hierarchical structure of the Hybrid Manifold Architecture System.

Its primary purpose is to define how quantities belonging to different sectors combine, interact, and contribute to physical observables.

The fundamental operations of Obidi Calculus are:

1. Addition.
2. Multiplication.
3. Contraction.
4. Determinant Composition.
5. Measure Construction.
6. Functional Factorization.

Together these operations form the algebraic backbone of the Theory of Entropicity.

7.2 Fundamental Principle

The central principle underlying Obidi Calculus may be stated as follows:

Every physical quantity is composed of sectoral contributions whose interactions obey well-defined algebraic rules.

Symbolically,

$$\text{Physical Quantity} = \text{Sectoral Contributions} + \text{Sectoral Interactions}.$$

The objective of Obidi Calculus is to formalize these contributions and interactions.

Part I

The Obidi Addition Rule

7.3 Definition

The Obidi Addition Rule states that the total value of a physical quantity is obtained by summing its sectoral contributions.

For a generic tensor T ,

$$T = \sum_J T^{(J)}.$$

Since

$$J \in \{FR, FS, L\},$$

this becomes

$$T = T^{(FR)} + T^{(FS)} + T^{(L)}.$$

This is the most fundamental algebraic operation in ToE.

7.4 Addition of Scalars

For scalar quantities,

$$\Phi = \Phi^{(FR)} + \Phi^{(FS)} + \Phi^{(L)}.$$

Examples include:

$$R = R^{(FR)} + R^{(FS)} + R^{(L)},$$

$$\Lambda = \Lambda_{FR} + \Lambda_{FS} + \Lambda_L.$$

Thus scalar observables may receive contributions from multiple sectors.

7.5 Addition of Vectors

For vectors,

$$V_\mu = V_{(\mu)FR} + V_{(\mu)FS} + V_{(\mu)L}.$$

Each term represents the vector contribution originating within its corresponding sector.

The observable vector field is the sum of all contributions.

7.6 Addition of Rank-Two Tensors

For rank-two tensors,

$$T_{\mu\nu} = T_{(\mu\nu)FR} + T_{(\mu\nu)FS} + T_{(\mu\nu)L}.$$

The metric decomposition is therefore

$$g_{\mu\nu} = g_{\mu\nu}^{(FR)} + g_{\mu\nu}^{(FS)} + g_{\mu\nu}^{(L)}.$$

Similarly,

$$G_{\mu\nu} = G_{\mu\nu}^{(FR)} + G_{\mu\nu}^{(FS)} + G_{\mu\nu}^{(L)}.$$

7.7 Physical Interpretation

The Addition Rule expresses the principle that physical reality receives simultaneous contributions from:

- Classical information geometry.
- Quantum information geometry.
- Spacetime geometry.

The observable universe is therefore interpreted as the cumulative manifestation of all sectors.

Part II

The Obidi Multiplication Rule

7.8 Motivation

Addition combines sectoral contributions.

Multiplication combines sectoral structures.

Many important physical quantities are multiplicative rather than additive.

Examples include:

- Determinants.
- Partition functions.
- Path integrals.
- Probability amplitudes.
- Functional measures.

To accommodate these quantities, ToE introduces the Obidi Multiplication Rule.

7.9 Definition

For multiplicative quantities,

$$Q = \prod_J Q^{(J)}.$$

Explicitly,

$$Q = Q^{(FR)} Q^{(FS)} Q^{(L)}.$$

This operation represents sectoral factorization.

7.10 Determinant Factorization

The most important application concerns the metric determinant.

The HMAS determinant is defined by

$$|g| = |g_{FR}| |g_{FS}| |g_L|.$$

This relation is fundamental to the construction of the Local Obidi Action.

The corresponding volume element becomes

$$\sqrt{|g|} = \sqrt{|g_{FR}|} \sqrt{|g_{FS}|} \sqrt{|g_L|}.$$

7.11 Sectoral Volume Measure

The generalized HMAS measure is

$$dV_{HMAS} = \sqrt{|g_{FR}|} \sqrt{|g_{FS}|} \sqrt{|g_L|} d^4x.$$

This measure replaces the conventional spacetime volume element

$$\sqrt{|g|} d^4x$$

used in General Relativity.

The difference reflects the presence of informational geometry alongside spacetime geometry.

7.12 Partition Function Factorization

Let

$$Z$$

denote the partition function of the complete system.

The Obidi Multiplication Rule implies

$$Z_{HMAS} = \prod_J Z^{(J)}.$$

Therefore

$$Z_{HMAS} = Z_{FR} Z_{FS} Z_L.$$

This relation expresses the factorization of statistical contributions across sectors.

Part III

The Obidi Contraction Rule

7.13 Motivation

The most delicate operation in tensor calculus is contraction.

Because HMAS contains multiple sectors, unrestricted contractions would generate ambiguities and inconsistencies.

The Obidi Contraction Rule resolves this problem.

7.14 Formal Definition

Let

$$T^{(\mu)J}$$

and

$$S_{(\mu)K}$$

be tensors belonging to sectors J and K .

Then

$$T^{(\mu)J} S_{(\mu)K} = \delta_K^J T^{(\mu)J} S_{(\mu)J}.$$

This relation generalizes Einstein contraction to HMAS.

7.15 Sectoral Orthogonality

For distinct sectors,

$$J \neq K$$

one obtains

$$T^{(\mu)J} S_{(\mu)K} = 0.$$

This is the mathematical statement of Obidi Orthogonality.

Distinct sectors do not directly contract.

7.16 Examples

Fisher–Rao with Fisher–Rao

$$A^{(\mu)FR} B_{(\mu)FR} \neq 0. \quad =$$

Contraction is permitted.

Fubini–Study with Fubini–Study

$$A^{(\mu)FS} B_{(\mu)FS} \neq 0. \quad =$$

Contraction is permitted.

Lorentzian with Lorentzian

$$A^{(\mu)L} B_{(\mu)L} \neq 0. \quad =$$

Contraction is permitted.

Fisher–Rao with Lorentzian

$$A^{(\mu)FR} B_{(\mu)L} = 0.$$

Contraction is forbidden.

Part IV

Sectoral Curvature Algebra

7.17 Sectoral Ricci Scalars

Each sector possesses its own Ricci scalar:

$$\begin{aligned} R^{(FR)}, \\ R^{(FS)}, \\ R^{(L)}. \end{aligned}$$

The total curvature scalar is defined through the Addition Rule:

$$R = R^{(FR)} + R^{(FS)} + R^{(L)}.$$

7.18 Sectoral Einstein Tensors

Each sector possesses an Einstein tensor

$$G_{\mu\nu}^{(J)} = R_{\mu\nu}^{(J)} - \frac{1}{2}g_{\mu\nu}^{(J)}R^{(J)}.$$

The total Einstein tensor becomes

$$G_{\mu\nu} = \sum_J G_{\mu\nu}^{(J)}.$$

Part V

Entropic Algebra

7.19 Entropy Field Decomposition

The entropy field may itself be decomposed into sectoral contributions:

$$S = S_{FR} + S_{FS} + S_L.$$

This decomposition permits independent entropic dynamics within each sector.

7.20 Entropic Gradient Structure

The gradient of entropy becomes

$$\nabla_\mu S = \nabla_\mu S_{FR} + \nabla_\mu S_{FS} + \nabla_\mu S_L.$$

The kinetic term therefore decomposes as

$$(\nabla S)^2 = \sum_J g^{(J)\mu\nu} \nabla_\mu^{(J)} S \nabla_\nu^{(J)} S.$$

This expression plays a central role in the Local Obidi Action.

Part VI

Algebra of the Local Obidi Action

7.21 Sectoral Action Contributions

Each sector contributes an independent action:

$$I_{LOA}^{(J)}.$$

The complete action becomes

$$I_{LOA} = \sum_J I_{LOA}^{(J)}.$$

Explicitly,

$$I_{LOA} = I_{FR} + I_{FS} + I_L.$$

7.22 Sectoral Functional Variation

Variation proceeds independently within each sector:

$$\delta I_{LOA} = \sum_J \delta I_{LOA}^{(J)}.$$

Consequently each sector generates its own field equations.

This property greatly simplifies the derivation of the Obidi Field Equations.

7.23 Closure Principle

A successful algebra must remain closed under its permitted operations.

Obidi Calculus satisfies this requirement.

Addition, multiplication, contraction, and variation all produce quantities that remain within HMAS.

No operation generates objects outside the theory's mathematical framework.

This property ensures internal consistency.

7.24 The Mathematical Role of Obidi Calculus

The significance of Obidi Calculus extends beyond notation.

It provides the algebraic machinery required to:

- construct actions,
- define measures,
- derive field equations,
- compute observables,
- relate informational geometry to spacetime geometry.

In this sense, Obidi Calculus plays a role analogous to tensor calculus in General Relativity.

7.25 Transition to the Local Obidi Action

The Hybrid Manifold Architecture System provides the geometric foundation.

The Obidi Convention provides the indexing language.

Obidi Calculus provides the algebra.

These ingredients now permit the construction of the dynamical heart of the Theory of Entropicity:

the **Local Obidi Action (LOA)**.

The next chapter develops the Local Obidi Action from first principles, derives its structure sector by sector, and establishes the variational framework that ultimately leads to the Obidi Field Equations.